The Ephemeris

September 2020
Volume 31 Number 03- The Official Publication of the San Jose Astronomical Association

September 2020 - December 2020 Events
Schedule subject to change due to Covid-19 restrictions.
Please check our website for up to date information.

Board & Speaker Talk (online)
Saturday 10/3, 11/7, 12/5
Board Meetings: 6 -7:30pm
Speaker Talk: 8:00-9:30pm

Fix-It Day (2-4pm)
Currently on hold due to COVID-19 Shelter in Place

Solar Sunday (2-4pm)
Sunday 9/13, 10/4, 11/8, 12/6

Intro to the Night Sky Class (7:30-8:30pm)
9/25

Houge Park 1Q and 3Q In-Town Star Party
Currently on hold due to COVID-19 Shelter in Place

RCDO Starry Nights Star Party
Currently on hold due to COVID-19 Shelter in Place

Imaging SIG Mtg
Tuesday 9/15, 10/20, 11/17, 12/15

Astro Imaging Workshops at Little Uvas OSP
Currently on hold due to COVID-19 Shelter in Place

Armchair Star Party Online
Dates TBA

Fall Swap Meet
Sunday 11/22 - tentative

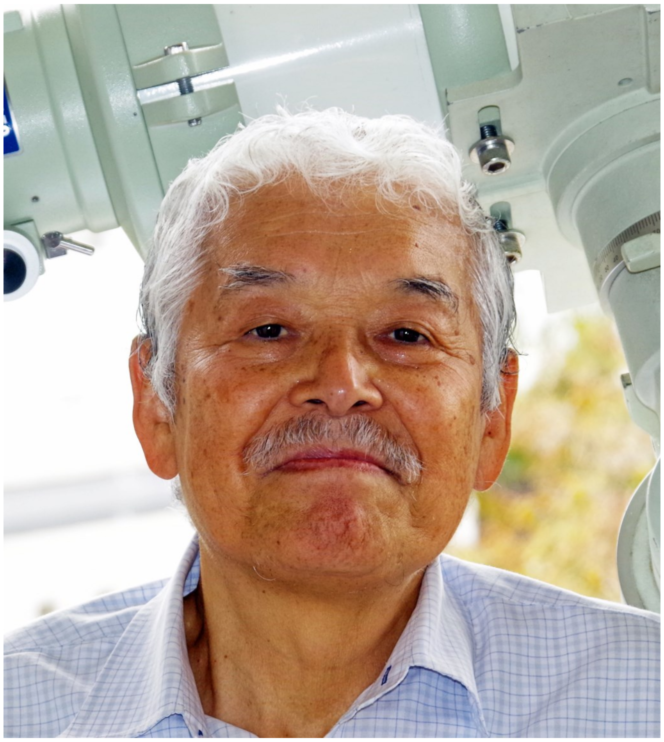
Please refer to the SJAA Website home page (www.sjaa.net) and click on any "Upcoming Event" highlighted in Blue. It will take you to the SJ-Astronomy Meetup page (www.meetup.com/SJ-Astronomy/events/) where you can find specific event times, locations, maps and possible cancellation due to weather.

INSIDE THIS ISSUE

Photo by SJAA Member: Hy Murveit
Oct. 20, 2019
Location: Ladera, CA

Greetings from the President

Credit: Ken Miura



Ken Miura in front of Takahashi TOA150 In Tokyo, Japan

Hello SJAA Members,

My name is Kenichi Miura (people usually call me "Ken"). I was elected as the new president of SJAA in August. Before becoming the president, I had served as Vice President of the club since March.

I first became a member of the club in 1988. My background is in experimental physics and computer science. I have worked at Fujitsu, a Japanese computer company, for more than 30 years, mostly involved in the high-end mainframes and supercomputers. My professional experience with astronomy includes a joint project with the National Astronomical Observatory of Japan (NAOJ) in the early 1980's. I was the chief architect at Fujitsu to design a special purpose system called FX(F stands for Fast Fourier Transform and X stands for cross-correlation) for radio astronomy. A modern version of this system is currently operational in the ALMA Project in Chile.

Since the time I became an officer of SJAA, the club has been having a difficult time with COVID-19, canceling or suspending all the in-person activities. Building 1 of Hogue Park has been closed by an order from the City of San Jose. At the same time, with the efforts of capable and enthusiastic club members, we have successfully explored a new direction toward virtual and on-line observations. Some good examples are the "Armchair Star Party" and "Online Solar Observation". All the club meetings are also shifting to an online format, such as the board meeting and the imaging SIG meetings.

As for the recent astronomical events, the biggest topic was by far the arrival of the NEOWISE comet in July. With the good weather, many members have observed and photographed it in the morning sky, and later, in the evening sky. There are some beautiful photos of the comet in this issue of Ephemeris too. Jupiter and Saturn brightly shining next to each other near the Milky Way is also a fantastic view. We are also expecting the approach of Mars in October. Its apparent size will exceed 20 arcseconds!

I hope the situation regarding COVID-19 will improve before too long to allow the in-person activities again; at the same time, the new direction toward online astronomy will continue to grow as part of the SJAA's activities. Since March, we have been discussing the strategic planning for future directions. You will hear from us about details in the near future, so stay tuned!

Regards,
Ken Miura

In Memoriam to Marilyn Perry

Credit: Jonathan Perry

Marilyn with her telescope at SJAA Pinnacles Star Party

Credits: Gary Chock

Yosemite Star Party Glacier Point

Credits: Jonathan Perry

As we are all aware, our long term member and friend Marilyn Perry passed away recently. Marilyn was born on July 26th, 1944 in Columbus Ohio and passed away on July 4th 2020. She was 75 years old. Marilyn is survived by her husband of 51 years Jonathan, their son James, and by older sister Barbara Hare and younger brother Nolan Doesken.

Below are some thoughts from some of her friends from SJAA

"I will always have fond memories of our friendship enriched by our shared observing experiences, her enthusiasm as she learned visual observing techniques, and joy in finding her Astronomy League list goals. "
- Gary Chock

"What a loss, Marilyn was always so encouraging with the school Star Party program and I am delighted in hearing her share her discoveries. The "merit badges" are perfect and what a showcase of her infectious curiosity."
- Cat Cvengros (School Star Party)

"I met Marilyn at many events - Starry Nights and ITSP and also some off-routine events that she volunteered for. She did love to share her knowledge and I have benefited many times from her experience with star gazing and her willingness to share the views with everybody around her. She will definitely be missed."
- Swami Nigam

"Her smile, energy, passion for the hobby, love for all the folks in the club or hobby or having some interest in Astronomy or visitors to our in-person events will be unbeatable! We are definitely going to miss her!!! Deepest condolences to her husband, family, and friends!"
- Rashi G.

"I am heart-broken. Marilyn was a kind, patient, knowledgeable, loving mentor to me. I admired her so much. I hadn't seen her for a while and was missing her. I was so glad to see her Southern Skies Observation Report in our last Ephemeris. Yes, Rashi, I think that is a wonderful idea to dedicate the next Armchair Star Party in her memory and honor."
- Marianne

"Susan & I are so sorry to hear of her passing. She was an enthusiastic, very knowledgeable astronomer, and a great 'senior' role model to encourage young women to get into STEM activities like astronomy. Marilyn loved helping with the annual "Stanford Tech Trek" that invited 80+ girls to a week-long STEM program at Stanford. SJAA 'women' and other astronomers from the Bay area would put on a Star Party for the girls."
- Bill and Susan O'Neil

NASA's Parker Solar Probe Spies Newly-Discovered Comet NEOWISE

Credit: NASA

An unprocessed image from the WISPR instrument on board NASA's Parker Solar Probe shows comet NEOWISE on July 5, 2020, shortly after its closest approach to the Sun. The Sun is out of frame to the left. The faint grid pattern near the center of the image is an artifact of the way the image is created. The small black structure near the lower left of the image is caused by a grain of dust resting on the imager's lens.

Credits: NASA/Johns Hopkins APL/Naval Research Lab/Parker Solar Probe/Brendan Gallagher

Processed data from the WISPR instrument on NASA's Parker Solar Probe shows greater detail in the twin tails of comet NEOWISE, as seen on July 5, 2020. The lower, broader tail is the comet's dust tail, while the thinner, upper tail is the comet's ion tail.

Credits: NASA/Johns Hopkins APL/Naval Research Lab/Parker Solar Probe/Guillermo Stenborg

NASA's Parker Solar Probe was at the right place at the right time to capture a unique view of comet NEOWISE on July 5, 2020. Parker Solar Probe's position in space gave the spacecraft an unmatched view of the comet's twin tails when it was particularly active just after its closest approach to the Sun, called perihelion. ASA's Parker Solar Probe was at the right place at the right time to capture a unique view of comet NEOWISE on July 5, 2020. Parker Solar Probe's position in space gave the spacecraft an unmatched view of the comet's twin tails when it was particularly active just after its closest approach to the Sun, called perihelion.

The comet was discovered by NASA's Near-Earth Object Wide-field Infrared Survey Explorer, or NEOWISE, on March 27. Since then, the comet — called comet C/2020 F3 NEOWISE and nicknamed comet NEOWISE — has been spotted by several NASA spacecraft, including Parker Solar Probe, NASA's Solar and Terrestrial Relations Observatory, the ESA/NASA Solar and Heliospheric Observatory, and astronauts aboard the International Space Station.

The image above is unprocessed data from Parker Solar Probe's WISPR instrument, which takes images of the Sun's outer atmosphere and solar wind in visible light. WISPR's sensitivity also makes it well-suited to see fine detail in structures like comet tails. Parker Solar Probe collected science data through June 28 for its fifth solar flyby, but the availability of additional downlink time allowed the team to take extra images, including this image of comet NEOWISE.

The twin tails of comet NEOWISE are seen more clearly in this image from the WISPR instrument, which has been processed to increase contrast and remove excess brightness from scattered sunlight, revealing more detail in the comet tails.

The lower tail, which appears broad and fuzzy, is the dust tail of comet NEOWISE — created when dust lifts off the surface of the comet's nucleus and trails behind the comet in its orbit. Scientists hope to use WISPR's images to study the size of dust grains within the dust tail, as well as the rate at which the comet sheds dust.

The upper tail is the ion tail, which is made up of gases that have been ionized by losing electrons in the Sun's intense light. These ionized gases are buffeted by the solar wind — the Sun's constant outflow of magnetized material — creating the ion tail that extends directly away from the Sun. Parker Solar Probe's images appear to show a divide in the ion tail. This could mean that comet NEOWISE has two ion tails, in addition to its dust tail, though scientists would need more data and analysis to confirm this possibility.

NASA Astronauts Safely Splash Down after First Commercial Crew Flight to Space Station

Credit NASA

Two NASA astronauts splashed down safely in the Gulf of Mexico Sunday for the first time in a commercially built and operated American crew spacecraft, returning from the International Space Station to complete a test flight that marks a new era in human spaceflight.

SpaceX's Crew Dragon, carrying Robert Behnken and Douglas Hurley, splashed down under parachutes in the Gulf of Mexico off the coast of Pensacola, Florida at 2:48 p.m. EDT Sunday and was successfully recovered by SpaceX. After returning to shore, the astronauts immediately will fly back to Houston.

"Welcome home, Bob and Doug! Congratulations to the NASA and SpaceX teams for the incredible work to make this test flight possible," said NASA Administrator Jim Bridenstine. "It's a testament to what we can accomplish when we work together to do something once thought impossible. Partners are key to how we go farther than ever before and take the next steps on daring missions to the Moon and Mars."

Behnken and Hurley's return was the first splashdown for American astronauts since Thomas Stafford, Vance Brand, and Donald "Deke" Slayton landed in the Pacific Ocean off the coast of Hawaii on July 24, 1975, at the end of the Apollo-Soyuz Test Project.

NASA's SpaceX Demo-2 test flight launched May 30 from the Kennedy Space Center in Florida. After reaching orbit, Behnken and Hurley named their Crew Dragon spacecraft "Endeavour" as a tribute to the first space shuttle each astronaut had flown aboard.

Nearly 19 hours later, Crew Dragon docked to the forward port of the International Space Station's Harmony module May 31.

"On behalf of all SpaceX employees, thank you to NASA for the opportunity to return human spaceflight to the United States by flying NASA astronauts Bob Behnken and Doug Hurley," said SpaceX President and Chief Operating Officer Gwynne Shotwell. "Congratulations to the entire SpaceX and NASA team on such an extraordinary mission. We could not be more proud to see Bob and Doug safely back home—we all appreciate their dedication to this mission and helping us start the journey towards carrying people regularly to low Earth orbit and on to the Moon and Mars. And I really hope they enjoyed the ride!"

Behnken and Hurley participated in a number of scientific experiments, spacewalks and public engagement events during their 62 days aboard station. Overall, the astronaut duo spent 64 days in orbit, completed 1,024 orbits around Earth and traveled 27,147,284 statute miles.

The astronauts contributed more than 100 hours of time to supporting the orbiting laboratory's investigations. Hurley conducted the Droplet Formation Study inside of the Microgravity Science Glovebox (MSG), which evaluates water droplet formation and water flow. Hurley also conducted the Capillary Structures investigation, which studies the use of different structures and containers to manage fluids and gases.

Hurley and Behnken worked on numerous sample switch outs for the Electrolysis Measurement (EM) experiment, which looks at bubbles created using electrolysis and has implications for numerous electrochemical reactions and devices. Both crew members also contributed images to the Crew Earth Observations (CEO) study. CEO images help record how our planet is changing over time, from human-caused changes – such as urban growth and reservoir construction – to natural dynamic events, including hurricanes, floods, and volcanic eruptions.

Behnken conducted four spacewalks while on board the space station with Expedition 63 Commander and NASA colleague Chris Cassidy. The duo upgraded two power channels on the far starboard side of the station's truss with new lithium-ion batteries. They also routed power and Ethernet cables, removed H-fixtures that were used for ground processing of the solar arrays prior to their launch, installed a protective storage unit for robotic operations, and removed shields and coverings in preparation for the arrival later this year of the Nanoracks commercial airlock on a SpaceX cargo delivery mission.

Continued on Page 6

Behnken now is tied for most spacewalks by an American astronaut with Michael Lopez-Alegria, Peggy Whitson, and Chris Cassidy, each of whom has completed 10 spacewalks. Behnken now has spent a total of 61 hours and 10 minutes spacewalking, which makes him the U.S. astronaut with the third most total time spacewalking, behind Lopez-Alegria and Andrew Feustel, and the fourth most overall.

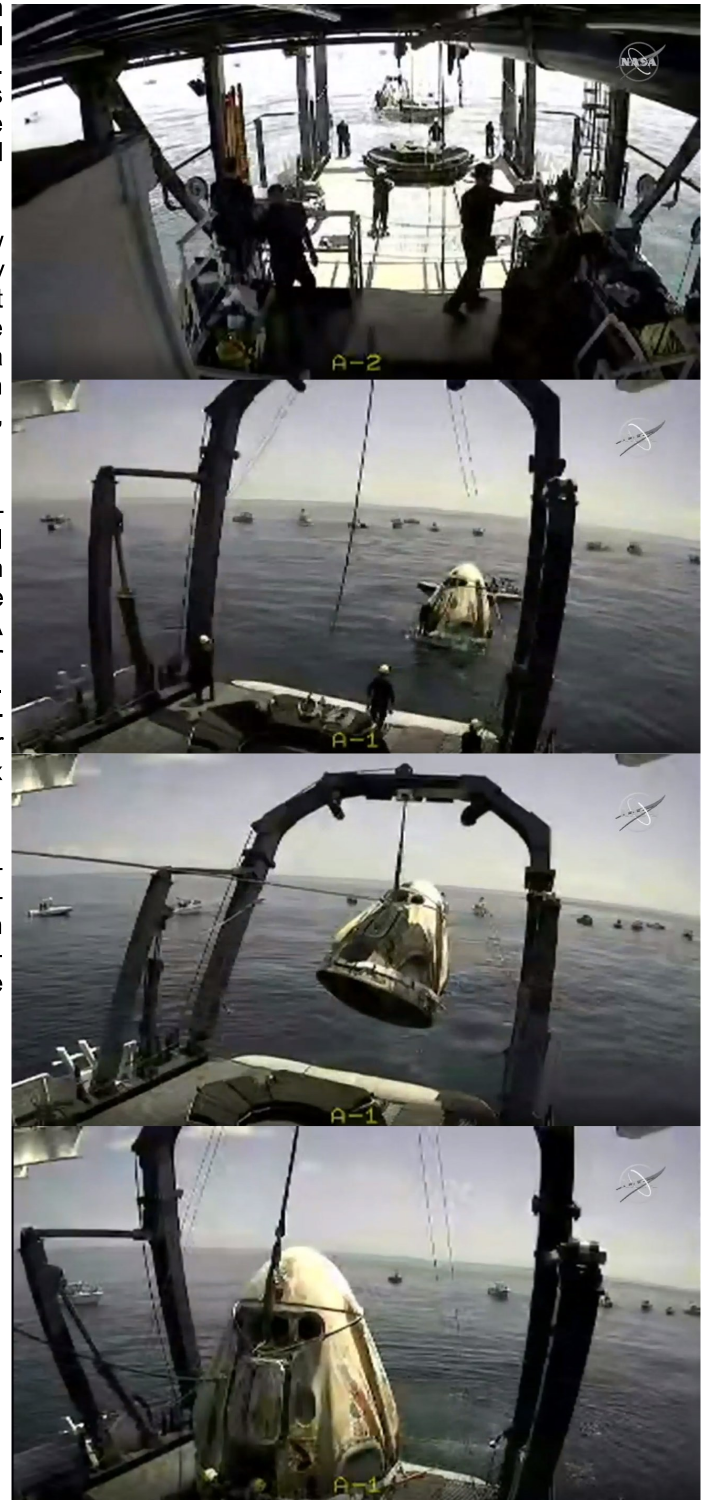
The Demo-2 test flight is part of NASA’s Commercial Crew Program, which has worked with the U.S. aerospace industry to launch astronauts on American rockets and spacecraft from American soil to the space station for the first time since 2011. This is SpaceX’s final test flight and is providing data on the performance of the Falcon 9 rocket, Crew Dragon spacecraft and ground systems, as well as in-orbit, docking, splashdown, and recovery operations.

Crew Dragon Endeavour will return back to SpaceX’s Dragon Lair in Florida for inspection and processing. Teams will examine the spacecraft’s data and performance from throughout the test flight. The completion of Demo-2 and the review of the mission and spacecraft pave the way for NASA to certify SpaceX’s crew transportation system for regular flights carrying astronauts to and from the space station. SpaceX is readying the hardware for the first rotational mission, called Crew-1, later this year. This mission would occur after NASA certification, which is expected to take about six weeks.

The goal of NASA’s Commercial Crew Program is safe, reliable and cost-effective transportation to and from the International Space Station. This could allow for additional research time and increase the opportunity for discovery aboard humanity’s testbed for exploration, including helping us prepare for human exploration of the Moon and Mars.

NASA astronauts Robert Behnken, left, and Douglas Hurley are seen inside the SpaceX Crew Dragon Endeavour spacecraft onboard the SpaceX GO Navigator recovery ship shortly after having landed in the Gulf of Mexico off the coast of Pensacola, Florida, Sunday, Aug. 2, 2020.

Credits: NASA/Bill Ingalls



SpaceX’s Crew Dragon, carrying NASA astronauts Robert Behnken and Douglas Hurley, splashes down in the Gulf of Mexico off the coast of Pensacola, Florida at 2:48 p.m. EDT Aug. 2, 2020, where the spacecraft is recovered by SpaceX and brought aboard the recovery ship 'Go Navigator.

Credits: NASA Television

SJAA Member Astrophoto Gallery

Date: August 15, 2019

Location: Little Uvas Morgan Hill, Ca

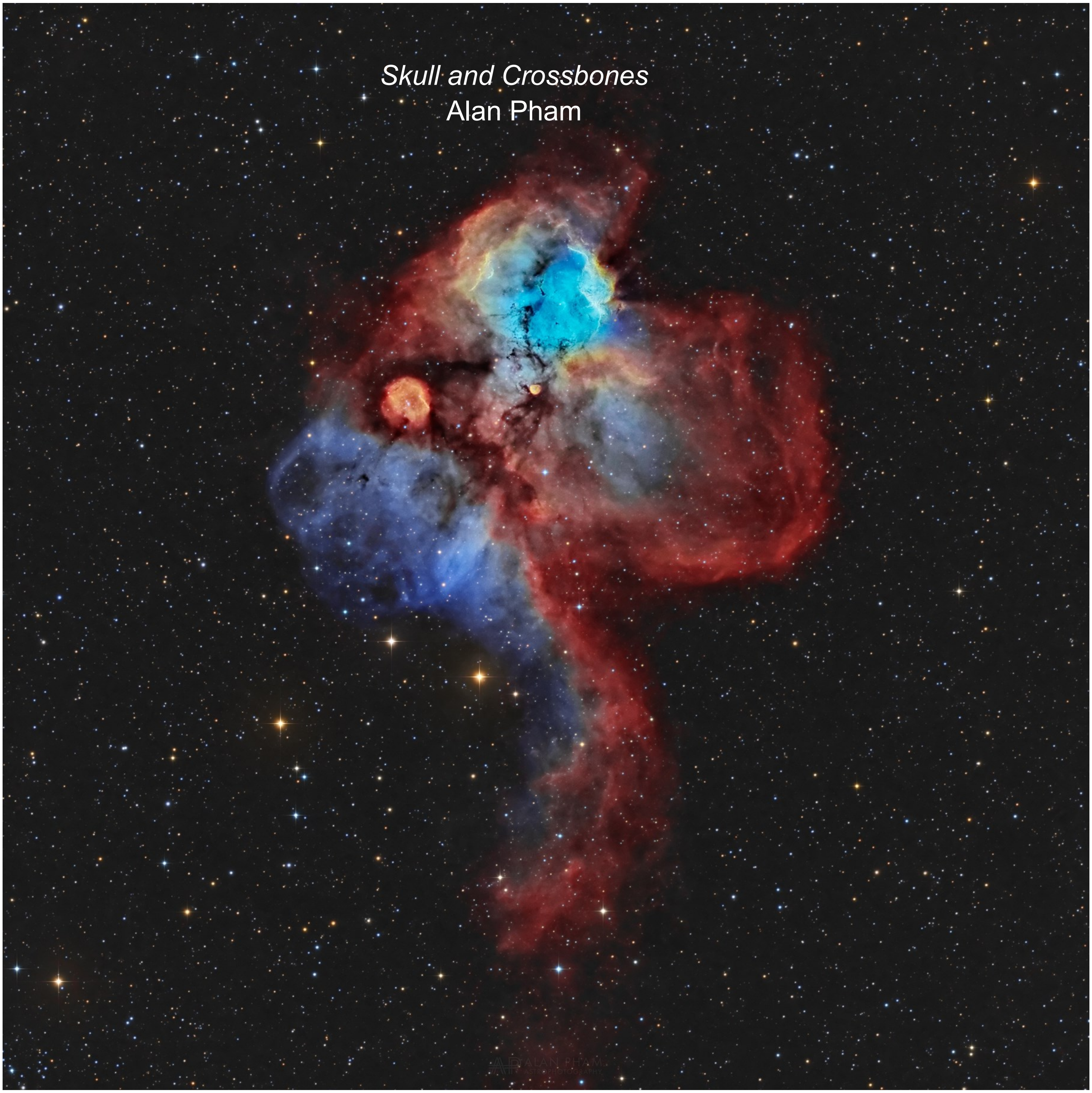
Imaging Telescope: Nikon 18-55mm Lens

Imaging Camera: Nikon D 5600

Mount: n/a

Software: Adobe Photoshop CC (Cloud)

SJAA Member Astrophoto Gallery



Dates: Feb. 14, 2020 , Feb. 18, 2020 , March 30, 2020 , April 2, 2020 , April 13, 2020

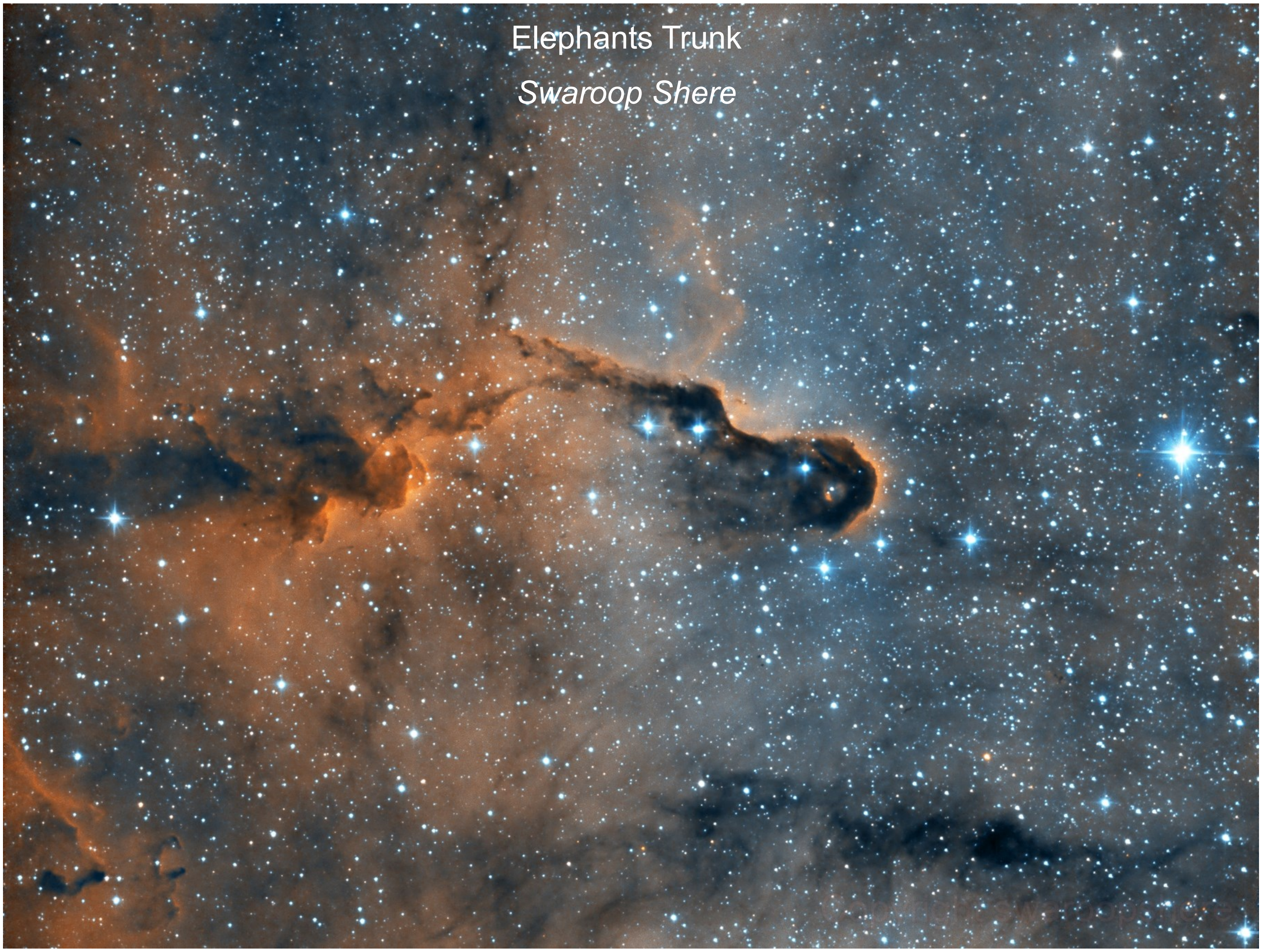
Location: San Jose, California, United States

Telescope: Stellarvue SV4

Camera: ZWO ASI1600MM-Cool

Mount: Astro-Physics Mach1 GTO

Software: PhotoShop , PHD2 , Sequence Generator Pro , PixInsight



Date: September 02, 2018

Location: Stellar Skies Observatory, Pontotoc, Tx, United States

Telescope: Orion 10" f/3.9 Astrograph

Camera: Starlight Express Lodestar X2

Mount: Software Bisque Paramount Paramount MYT

Software: Software Bisque Sky X Pro , PixInsight 1.8 PixInsight , Seqence Generator Pro , PHD2 Guiding

SJAA Member Astrophoto Gallery

Dates: June 6, 2020

Location: Sierra remote observatories

Telescope: FSQ-106ED

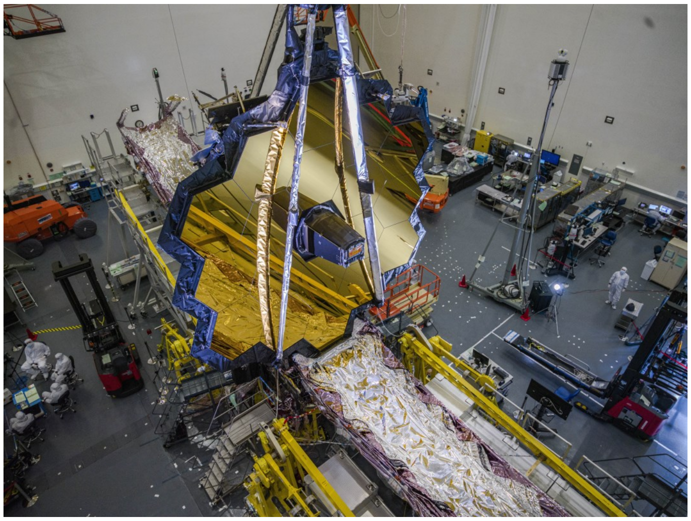
Camera: Lodestar

Mount: Paramount ME

Software: PixInsight

NASA Announces New James Webb Space Telescope Target Launch Date

Credit: NASA



NASA's James Webb Space Telescope in the clean room at Northrop Grumman, Redondo Beach, California, in July 2020 .

Credits: NASA/Chris Gunn

NASA now is targeting Oct. 31, 2021, for the launch of the agency's James Webb Space Telescope from French Guiana, due to impacts from the ongoing coronavirus (COVID-19) pandemic, as well as technical challenges.

This decision is based on a recently completed schedule risk assessment of the remaining integration and test activities prior to launch. Previously, Webb was targeted to launch in March 2021.

"The perseverance and innovation of the entire Webb Telescope team has enabled us to work through challenging situations we could not have foreseen on our path to launch this unprecedented mission," said Thomas Zurbuchen, associate administrator for NASA's Science Mission Directorate at the agency's headquarters in Washington. "Webb is the world's most complex space observatory, and our top science priority, and we've worked hard to keep progress moving during the pandemic. The team continues to be focused on reaching milestones and arriving at the technical solutions that will see us through to this new launch date next year."

Testing of the observatory continues to go well at Northrop Grumman, the mission's main industry partner, in Redondo Beach, California, despite the challenges of the coronavirus pandemic. Prior to the pandemic's associated delays, the team made significant progress in achieving important milestones to prepare for launch in 2021.

As schedule margins grew tighter last fall, the agency planned to assess the progress of the project in April. This assessment was postponed due to the pandemic and was completed this week. The factors contributing to the decision to move the launch date include the impacts of augmented safety precautions, reduced on-site personnel, disruption to shift work, and other technical challenges. Webb will use existing program funding to stay within its $8.8 billion development cost cap.

"Based on current projections, the program expects to complete the remaining work within the new schedule without requiring additional funds," said Gregory Robinson, NASA Webb program director at the agency's headquarters. "Although efficiency has been affected and there are challenges ahead, we have retired significant risk through the achievements and good schedule performance over the past year. After resuming full operations to prepare for upcoming final observatory system-level environmental testing this summer, major progress continues towards preparing this highly complex observatory for launch."

The project team will continue to complete a final set of extremely difficult environmental tests of the full observatory before it will be shipped to the launch site in Kourou, French Guiana, situated on the northeastern coast of South America.

"Webb is designed to build upon the incredible legacies of the Hubble and Spitzer space telescopes, by observing the infrared universe and exploring every phase of cosmic history," said Eric Smith, NASA Webb's program scientist at the agency's headquarters. "The observatory will detect light from the first generation of galaxies that formed in the early universe after the big bang and study the atmospheres of nearby exoplanets for possible signs of habitability."

Early next year, Webb will be will folded "origami-style" for shipment to the launch site and fitted compactly inside Arianespace's Ariane 5 launch vehicle fairing, which is about 16 feet (5 meters) wide. On its journey to space, Webb will be the first mission to complete an intricate and technically challenging series of deployments – a critical part of Webb's journey to its orbit about one million miles from Earth. Once in orbit, Webb will unfold its delicate five-layered sunshield until it reaches the size of a tennis court. Webb will then deploy its iconic 6.5-meter primary mirror that will detect the faint light of far-away stars and galaxies.

Kid Spot

Kid Spot Jokes:

- **How do Mars and Jupiter keep their pants up ?**
  With their asteroid belt.
- **What's a light-year?**
  The same as a regular year, but with less calories.
- **What happened to the astronaut who stepped on chewing gum?**
  He got stuck in an Orbit!

Fun Facts

Comet NEOWISE

Comet NEOWISE taken at Pinnacles National Park

Credit: Nikola Nikolov

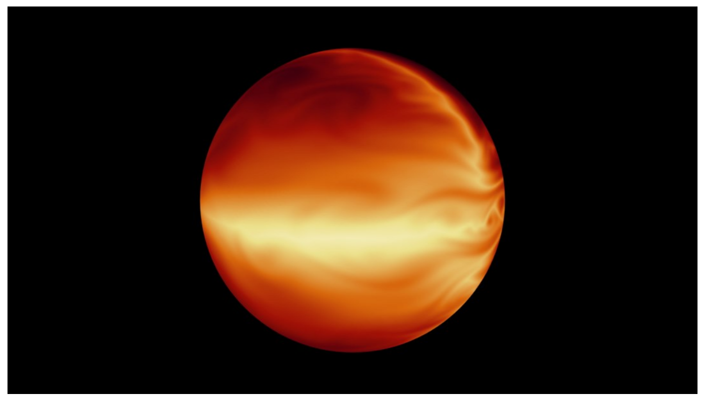
NEOWISE (C/2020 F3) is a comet, and similar to other comets, it has a really long orbital period around the Sun, when compared to our planet earth. Earth goes around the Sun once every year, but it takes 6,800 to 7,000 years for comet NEOWISE to go around the sun!

The closest comet NEOWISE will come to the earth is 64 million miles, which is about 270 times the distance to the moon!
Comet NEOWISE is travelling at about 40miles / second and is just about 3 miles in diameter

Comet NEOWISE could be observed with two tails:

* the first tail - blue and made of gas and ions stretching almost 70° from its nucleus (with a red separation in the tail caused by high amounts of sodium which is nearly stretched 1°),
* the second twin tail - golden color and made of dust stretched almost 50°, like the tail of Comet Hale–Bopp

Hot Jupiters



The turbulent atmosphere of a hot, gaseous planet known as HD 80606b is shown in this simulation based on data from NASA's Spitzer Space Telescope.

Credits: NASA/JPL-Caltech/MIT/Principia College

Hot Jupiters are physically very close to Jupiter, but have extremely short orbital periods, generally in the range of 0 -10 days! They are fundamentally a class of giant ex-oplanets.

They have very high surface-atmosphere temperatures due to the extreme proximity to their stars, and thus are termed Hot-Jupiters!

They are known to have very low densities, but the large radii of hot Jupiters are not yet fully understood!

The day-night temperature difference at the photosphere is predicted to be substantial, approximately 500 K (226 deg C)!!

They could be anywhere between 0.36–11.8 Jupiter masses. (They are pretty massive!!) but have low orbital periods ranging from 1.3–111 Earth days!Ultra-hot Jupi-ters are hot Jupiters with a dayside temperature greater than 2200K!!

From the Board of Directors

Announcements

- ◊ San Jose Astronomical Association is closely monitoring the coronavirus/COVID-19 situation in Santa Clara County. In accordance with guidance from Santa Clara County Health Department, SJAA leaders have been canceling many events, especially those that involve close contact and shared surfaces on telescopes. Some events, however, have successfully been moved to an online format. Please check our Meetup to determine if an event has been canceled or moved online. The board of directors will periodically review the status of the outbreak and the guidance from the county to determine the impact upon upcoming events.
- ◊ The board unanimously approved Ken Miura, current VP and acting President, to serve as the new President of SJAA for the remainder of Gerry's term, effective 8 Aug 2020. Consequently, the VP Position is now open and the President will commence a search for a new VP. For a note from the President please see page 2 of this edition.
- ◊ SJAA is also working on plans to restart the Loaner program, by conducting it within the guidelines of CDC and the county, and without allowing any public into the Houge Park. Any updates on this will be sent out over email.
- ◊ The National Park Service has decided to cancel all the summer activities and the entire summer program of next year due to park maintenance. The star party will be scheduled to resume in 2022.

Board Meeting Excerpts

July 11, 2020

In attendance
Swami, Sandy, Emi, Glenn, Ken, Peter Wolf, Sukhada, Rob J, Vini
Excused Absences: Peter
Guests: Marianne, Rahul, Beth, Emre, Nancy, Rashi, Laurie M, Kanch, Carl Wong

Broader Advertising of On-Line Events (NASA NSN)
Wolf suggested that our on-line events not be limited to the Bay Area (i.e. our normal Meetup group membership). SJAA's July events have been put up on NASA's Night Sky Network (NSN).

Social media and publicity for SJAA
SJAA has multiple social media outlets - Twitter, Instagram, Facebook. A new member, Beth Johnson, has prior experience managing social media for other organizations and has volunteered to manage all of SJAA social media outlets and make them more active. A public relations component might be added to this effort to interact with any external media outreach or interactions, such as with KCBS last year.

YouTube Channel for SJAA
A new YouTube channel under the name - "SJ-Astronomy" has been created. All relevant SJAA online events will be streamed there. It will also serve as a repository for such recordings.

August 8, 2020

In attendance
Swami, Sandy, Vini, Glenn, Sukhada, Wolf , Ken, Emi
Excused Absences: Rob J, Peter
Guests: Marianne, Kanch, Rashi, Rahul

New program for Electronically Assisted Astronomy
With SJAA's shift to doing many online programs such as Armchair Star Party, Intro to Night sky, Solar, etc., a new dedicated program for Electronically Assisted Astronomy will possibly be added, and will include online public events and digital material to back them up. The proposed charter is being updated based on the feedback from the board members, and the updated proposal will be presented at the next board meeting.

Ephemeris Editor positions
As per the bylaws, the editors are appointed by the President. Swami suggests that since a new President takes over in March, it would be best for the President to make his/her appointments at the following June board meeting, after the June edition has been published.

Astronomy Hangout Events
Due to the limitations of YouTube for active interactions, a pilot is planned to be run on Google Meet or Zoom, where people can call in for an interactive chat. The session is to have an attendance limit of 16 - 20 people, 2-3 SJAA hosts, with Q&A.

Swap Meet
Spring swap meet was canceled due to COVID-19. Next, swap meet is the fall which is tentatively scheduled for November 22 subject to the Covid-19 situation and restrictions in place at that time.

August 29, 2020

In attendance
Swami, Wolf, Vini, Rob J, Glenn, Emi, Ken, Swami, Peter, Sandy, Sukhada
Excused Absences: None
Guests: Marianne, Kanch, Dave I

Loaner Program
The SJAA Loaner Program receives requests from existing members, and specially new members on loaner equipment availability. The proposal will be updated based on the discussion and resubmitted to the board for approval. SJAA will reach out to the city for approval to use Houge Park for this event while ensuring Covid-19 related safety measures

24" Dobb sale / donation
SJAA plans to sell / donate a 24" Dobsonian mount telescope, and is open to SJAA members' suggestions on its use. The scope takes 2 experienced members to set it up and move it around. The board is open to suggestions from members on how best to utilize it. If there are no ideas about how the club can utilize it, the club will then consider selling it.

SJAA Library

SJAA offers another wonderful resource; a library with good astronomy books and DVDs available to all of our members that will interest all age groups and especially young children who are budding astronomers! You may still send all your questions/comments to librarian@sjaa.net.

Telescope Fix It Session

Fix It Day, sometimes called the Telescope Tune Up or the Telescope Fix It program is a real simple service the SJAA offers to members of the community for free, though it's priceless. Headed up by Vini Carter, the Fix It session provides a place for people to come with their telescope or other astronomy gear problems and have them looked at, such as broken scopes whose owners need advice, or need help with collimating a telescope.

http://www.sjaa.net/programs/fixit/

Solar Observing

Solar observing sessions, headed up by Wolf Witt, are usually held the 1st Sunday of every Month from 2pm - 4pm at Houge Park weather permitting. Please check SJ Astronomy Meetup for schedule details as the event time / location is subject to change.

http://www.meetup.com/SJ-Astronomy/

Quick STARt Program

The Quick STARt Program, headed up by Dave Ittner, helps to ease folks into amateur astronomy. You have to admit, astronomy can look exciting from the outside, but once you scratch the surface, it can get seemingly complex in a hurry. But it doesn't have to be that way if there's someone to guide you and answer all your seemingly basic questions.

The Quick STARt program is undergoing a redesign to make it more accessible to SJAA members, so signups are currently on hold.

http://www.sjaa.net/programs/quick-start/

Intro to the Night Sky

The Intro to the Night Sky session takes place monthly, in conjunction with first quarter moon and In Town Star Parties at Houge Park. This is a regular, monthly session, run by David Grover, each with a similar format, with only the content changing to reflect what's currently in the night sky. After the session, the attendees will go outside for a guided, green laser tour of the sky, along with views through SJAA Telescopes at the In-Town Star Party, to get a better look at the night's celestial objects.

http://www.sjaa.net/programs/beginners-astronomy/

Loaner Program

Muditha Kanchana (Kanch) heads up this program. The Program goal is for SJAA members to be able to evaluate equipment they are considering purchasing or are just curious about by checking out loaners from SJAA's growing list of equipment. Please note that certain items have restrictions or special conditions that must be met.
If you are an SJAA member and an experienced observer or have been through the SJAA Quick STARt program please fill this form to request a particular item. Please also consider donating unused equipment.
http://www.sjaa.net/programs/loaner-telescope-program/

Astro Imaging Special Interest Group (SIG)

SIG has a mission of bringing together people who have an interest in astronomy imaging, or put more simply, taking pictures of the night sky. Run by Bruce Braunstein, the Imaging SIG meets roughly every month at Houge Park to discuss topics about imaging. The SIG is open to people with absolutely no experience but want to learn what it's all about, but experienced imagers are also more than welcome, indeed, encouraged to participate. The best way to get involved is to review the postings on the SJAA Astro Imaging mail list in Google Groups.

http://www.sjaa.net/programs/imaging-sig/

Astro Imaging Workshops and Field Clinics

Not to be confused with the SIG group this newly organized program championed by Glenn Newell is a hands on program for club members, who are interested in astrophotography, to have a chance of seeing what it is all about.
Workshops are held at Houge Park once per month and field clinics (members only) once per quarter at a dark sky site. Check the schedule and contact Glenn Newell if you are interested.

School Star Party

The San Jose Astronomical Association conducts evening observing sessions (commonly called "star parties") for schools in mid-Santa Clara County, generally from Sunnyvale to Fremont to Morgan Hill.

Contact SJAA's (Program Coordinator) for additional information.

http://www.sjaa.net/programs/school-star-party/

SJAA Contacts

President/Dir.	Ken Miura
Vice-President:	Vacant
Treasurer/Dir:	Rob Jaworski
Secretary:	Emi Nikolov
Director:	Vini Carter
Director:	Sandy Mohan
Director:	Wolf Witt
Director:	Swami Nigam
Director:	Peter Melhus
Director:	Glenn Newell
Director:	Sukhada Palav
Ephemeris Newsletter -	
Content Editor:	Abishek Rama-subramanian
Prod. Editor:	Emi Nikolov
Binocular Stargazing:	Ed Wong
Fix-it Program:	Vini Carter
Hands on Imaging:	Glenn Newell
Imaging SIG:	Bruce Braunstein
Intro to the Night Sky:	David Grover
In-Town Star Party:	Muditha Kanchana
Library:	Jai Purandare
Loaner Program:	Muditha Kanchana
Memberships:	Kaushal Gangakhedkar
Publicity:	Rob Jaworski
Quick STARt	Dave Ittner
Donations:	Dave Ittner*
Solar:	Wolf Witt
Starry Nights:	Carl Wong
School Events:	Cat Cvengros
Refreshments:	Marianne Damon
Social Media Manager:	Jessica Johnson
Speakers:	Sukhada Palav
Webmaster:	Satish Vellanki

E-mails: http://www.sjaa.net/contact

SJAA Ephemeris, the newsletter of the San Jose Astronomical Association, is published quarterly.

Articles, including Observation Reports for publication should be submitted by no later than the 20th of the month of February, May, August and November.

San Jose Astronomical Association
P.O. Box 28243
San Jose, CA 95159-8243
http://www.sjaa.net/contact

San Jose Astronomical Association
P.O. Box 28243
San Jose, CA 95159-8243

Place
postage
here

Fold here

San Jose Astronomical Association Annual Membership Form

P.O. Box 28243 San Jose, CA 95159-8243

Membership Type (must be 18 years or older):

☐ **New** ☐ **Renewal**

☐ **$20** Regular Membership with **online** Ephemeris

☐ **$30** Regular Membership with **hardcopy** Ephemeris mailed to below address

The newsletter is always available online at: **http://www.sjaa.net/sjaa-newsletter-ephemeris/**

Questions? Send e-mail to: **memberships@sjaa.net**

Bring this form to any SJAA Meeting or send to the address (above). Make checks payable to "SJAA", or join/renew at: **http://www.sjaa.net/join-the-sjaa/**

Name: ______________________________

Address: ______________________________

City/ST/Zip: ______________________________

Phone: ______________________________

E-mail address: ______________________________